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# ④ Long-Tailed Distributions

## (1. The Myth of Normality)

↳ Reality check: Most real-world data (incomes, stock returns, web traffic) isn't normally distributed.

↳ Key Features of Long-Tailed Data.

→ Extreme values ("black swans") occur more frequently than predicted by normal distribution.

→ Asymmetric (skewed) or symmetric but with heavy tails.

Examples

The "Pizza Delivery Time" Surprise.

1. Normal Distribution Prediction

Imagine a pizza shop claims:

↳ Average delivery time: 30 minutes.

↳ Standard deviation: 5 minutes.

↳ Normal distribution assumptions:

- 68% of pizzas arrive in 25-35 mins.
- 95% ~~of~~ in 20-40 mins.
- 99.7% in 15-45 mins.

Prediction:

"2 hours (120 minutes) delivery should happen once every 1.4 million years (Based on normal distribution probabilities)

2. Reality (Long-Tailed Distribution).

Actual delivery times:

- Most pizzas arrive in 25-40 mins.
- But occasionally
  - ↳ Flat tires
  - ↳ wrong addresses
  - ↳ kitchen delays.

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Result

↳ 2-hour delays actually happen once every 3 months.

↳ The "tail" of extreme delays is MUCH fatter than predicted.

Key Takeaways

1) Normal distributions underestimate "rare" events in real life.

2) Long tails mean:

↳ Extreme values are more common than expected

↳ "Black swans" (like 2-hours delays) aren't really black swans means"

↓

This like means:

① Black Swan means:

-) Extremely rare, unpredictable events (like seeing a real black swan in nature).

-) Example: A meteor hitting your house.

② like is saying

-) Some things we call "black swans" (like 2-hour pizza delays) aren't actually that rare.

-) They happen regularly if you look at real data

-) We just pretend they're rare because our math models are wrong.

Real Pizza Examples:

- Bad math says: "2-hour delays should happen once every 100 years.
- Reality: They happen every Sunday when the Football game is on.

Key points:

Don't call something a "black swan" if it actually happens often - that's just bad planning

Simple Fix:

Look at your actual delivery history to see how often "rare" events really occur.

(2. Visualizing Long Tails.) watch code

(3 Types of Long-Tailed Distributions).

| Types | Descriptions | Examples | Python Generator |
|---|---|---|---|
| Power law | Extreme events scale exponentially | wealth, city population | np.random.pareto(a, size). |
| Log-Normal | Multiplicative processes | Income, stock prices | np.random.lognormal(μ, σ) |
| Cauchy | Heavy tails, no mean | Financial crashes | np.random.standard_cauchy() |
| Poisson | Counts of rare events. | website visits per second | np.random.poisson(lam, size). |

(4. Why it Matters)

Danger of Assuming Normality:

(1) Underesting Risk:

→ Normal models predict 0.3% chance of 4σ event.

→ Real finance data shows 5σ events every few years.

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2. Faulty Predictions:

→ Weather models assuming normality miss extreme storms

→ AI safety: Rare but catastrophic failures.

Better approaches

- Extreme Value Theory (EVT) for tail risk.
- Robust statistics (median/IQR instead of mean/$\sigma^2$)

5) Handling Long-Tailed Data.

(a) Transformations (b) Alternative Models

(c) Tail Risk Metrics.

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# Pizza Delivery Times Examples

```
Import numpy as np.
from scipy import stats.
import matplotlib.pyplot as plt

# Simulate pizza delivery times (in minutes).
# Most deliveries take 25-35 mins, but
  occasionally much longer.

np.random.seed(42).
normal_deliveries = np.random.normal(loc=30,
                                     scale=5, size=500).
long_tail_extremes = np.random.exponential
                     (scale=40, size=20)+30
                     # Some long delays.
delivery_time = np.concatenate([normal_
                deliveries, long_
                tail_extremes]).

# QQ plot to check normality
plt.figure(figsize=(10,6)).
stats.probplot(delivery_times, dist="norm",
               plot=plt)
plt.title("QQ-plot: Pizza Delivery times
          (Long Tails Visible)")
plt.title.show().
```